Lecture 8

Cyclic group: A group G is said to be cyclic if $\exists\, g \in G$ such that

$$G = \{g^n \mid n \in \mathbb{Z}\}$$

Theorem: A group of order n is cyclic if and only if it has an element of order n.

Proof: We have proved that if G has an element of order n, then G is cyclic.

Let G be a cyclic group of order n.

Let $G = \langle g \rangle = \{g^k \mid k \in \mathbb{Z}\}$.

We claim $|g| = n$ i.e. g is an element of order n.

First notice that $1, g, g^2, \dots, g^{m-1}$ are all distinct elements of G, where $m = |g|$.

If that is not true then there exists $0 \le i < j \le m-1$ such that

$$g^i = g^j \Rightarrow g^{j-i} = 1.$$

Contradicting $|g| = m$ as $j - i < m$.

So $1, g, g^2, \dots, g^{m-1}$ are all distinct.

$$\Rightarrow n = |G| \ge m.$$

as $G = \langle g \rangle = \{g^k \mid k \in \mathbb{Z}\}$

Now we claim $G \subset \{1, g, g^2, \dots, g^{m-1}\}$.

Let $x \in G$.

then $x = g^k$ for some $k \in \mathbb{Z}$.

By division algorithm $\exists$ unique integers q & r such that

$k = mq + r$. $\quad 0 \leq r < m$

Now $x = g^k = g^{mq+r}$

$= g^{mq} \cdot g^r$

$= (g^m)^q g^r$

$= (1)^q g^r = g^r \in \{1, g, g^2, \dots, g^{m-1}\}$

Hence $G \subset \{1, g, g^2, \dots, g^{m-1}\}$

So $G = \{1, g, g^2, \dots, g^{m-1}\}$

Hence $m = n$.

<u>Remark:</u> Let G be a group. & $g \in G$.

Define $H = \langle g^k \mid k \in \mathbb{Z} \}$.

Note $H \leq G$. H is a cyclic group of G generated by g.

we write $H = \langle g \rangle$.

Corollary: If G is a finite group, then the order of any element of G divides the order of G.

Proof: Let $g \in G$.

then $H = \langle g \rangle = \{g^k \mid k \in \mathbb{Z}\} \leq G$.

So by Lagrange's theorem.

$$|H| \mid |G|.$$

But $|H| = |g|$. [By previous theorem]

Hence $|g| \mid |G|$

Corollary: If G is a finite group & $g \in G$, then

$$g^{|G|} = 1.$$

Proof: We know $g^{|g|} = 1$.

& we also know that $|g| \mid |G|$

Hence $|G| = \ell |g|$

So $g^{|G|} = g^{\ell |g|} = (g^{|g|})^{\ell} = 1^{\ell} = 1$

As a Consequence of this, we can deduce Fermat's Little Theorem:

Fermat's Little Theorem: Let p be a prime number & a is any integer, Co-prime to p, then

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof: Consider the group $(\mathbb{Z}_p^*, \cdot)$.

$|\mathbb{Z}_p^*| = p-1$ as p is prime.

So $[a]^{p-1} = [1]$

$\Rightarrow [a^{p-1}] = [1]$

$\Rightarrow p \mid a^{p-1} - 1 \Rightarrow a^{p-1} \equiv 1 \pmod{p}$

Euler's Theorem: If n is a positive integer and a is another positive integer coprime to n, then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof: Consider the group $(\mathbb{Z}_n^*, \cdot)$

$|\mathbb{Z}_n^*| = \varphi(n)$

So $[a]^{\varphi(n)} = [1]$

$\Rightarrow [a^{\varphi(n)}] = [1]$

$\Rightarrow n \mid a^{\varphi(n)} - 1$

$\Rightarrow a^{\varphi(n)} \equiv 1 \pmod{n}$

Theorem: Let G be a group. Then

① $|a| = |gag^{-1}| \quad \forall a, g \in G$.

② If $g \in G$, $|g|$ is finite & $g^m = 1$ then $|g|$ divides m.

③ If $|g| = n$ then for $k \in \mathbb{Z}$,
$$|g^k| = \frac{n}{\gcd(k,n)}.$$

Proof: ① ~~Let $|a| = m$~~ To prove this we prove

$a^m = 1$ if and only if $(gag^{-1})^m = 1$.

Let $a^m = 1$.

then $(gag^{-1})^m = (gag^{-1})(gag^{-1}) \cdots (gag^{-1})$ (m times)
$$= ga^m g^{-1}$$
$$= g \cdot 1 g^{-1} = gg^{-1} = 1.$$

Conversely let
$$(gag^{-1})^m = 1.$$
$$\Rightarrow g a^m g^{-1} = 1$$
$$\Rightarrow g a^m = g$$
$$\Rightarrow a^m = g^{-1} g = 1. \quad \Rightarrow a^m = 1.$$

So $a^m = 1$ iff $(gag^{-1})^m = 1$

Hence $|a| = |gag^{-1}| \quad \forall a, g \in G$.

(2) Let $g \in G$. $|g| = n$.

Let $g^m = 1$.

Need to prove $n \mid m$.

By division algorithm, $\exists!$ q & r such that

$$m = nq + r \qquad 0 \leq r < n.$$

$$\Rightarrow 1 = g^m = g^{nr+r} = g^{nq} \cdot g^r$$

$$= (g^n)^q \cdot g^r = 1 \cdot g^r$$

$$\Rightarrow g^r = 1.$$

Since $0 \leq r < n$, $r = 0$. $\Rightarrow n \mid m$.

(3). We want to prove that

$$|g^k| = \frac{n}{\gcd(n,k)} \quad \text{where } |g| = n.$$

$$(g^k)^{\frac{n}{\gcd(n,k)}} = g^{\frac{kn}{\gcd(n,k)}} = (g^n)^{\frac{k}{\gcd(n,k)}} = 1.$$

By (2), $|g^k| \;\Big|\; \dfrac{n}{\gcd(n,k)}$.

Let $|g^k| = m$. $\Rightarrow (g^k)^m = 1$.

$$\Rightarrow g^{km} = 1.$$

$\Rightarrow |g| = n \mid km$. by (2).

$$\Rightarrow \frac{n}{\gcd(k,n)} \;\Big|\; \frac{k}{\gcd(k,n)} m.$$

As $\gcd\left(\frac{n}{\gcd(k,n)}, \frac{k}{\gcd(k,n)}\right) = 1$.

we have

$$\frac{n}{\gcd(k,n)} \mid m.$$

Hence $m = \frac{n}{\gcd(k,n)} = |g^k|$.

Corollary: Let G be a finite cyclic group. & let g be a generator of G i.e.

$$G = \{g^k \mid k \in \mathbb{Z}\}$$

then g^m is also a generator of G if $\gcd(m, |G|) = 1$.

Proof: Since g is a generator of G,

$$|g| = |G|.$$

$$|g^m| = \frac{|g|}{\gcd(m,|g|)} = \frac{|G|}{\gcd(m,|G|)}.$$

if $\gcd(m,|G|) = 1$ then $|g^m| = |G|$.

Hence g^m is also a generator of G.

Remark: # of generators of $G = \varphi(|G|)$.